BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: 25-26J-103

Project Title: E-Learning Platform for Hearing Impaired Students

<table>
<tr><td></td><td>Identification and Analysis</td></tr>
<tr><td>Data Privacy Concerns</td><td>Member 1 – Sign Language Synthesis with Question Interaction and Error Correction
The Sign Language Synthesis component processes three categories of sensitive educational data: lesson content-to-sign translation mappings, student-initiated question logs, and error correction histories. Unlike traditional learning management systems, this platform stores word-level sign mapping access patterns — recording exactly which vocabulary items, sentence structures, and lesson segments each student requested to be signed. A student who repeatedly requests sign translation for medical terminology, legal terms, or examination content reveals their specific areas of academic weakness or special interest.

Each query, when aggregated over time, creates a detailed cognitive load profile - revealing exactly which concepts the student struggles to comprehend through sign language alone. This constitutes sensitive educational metadata that could be used to infer learning disabilities beyond hearing impairment.
The Error Correction component stores:
Wrong answers with timestamps
Repeated mistake patterns
Failed activity attempts (up to 3 retries before teacher notification)
Correction response outcomes (whether the student understood after correction)
This creates a permanent remediation record that, if breached, could subject students to academic stigma or discriminatory treatment by future educational institutions.
Mitigations implemented:
All interaction logs stored with student_id hashed using SHA-256 with per-session salt - preventing cross-database correlation
Correction records automatically anonymized after 30 days, retaining only aggregated error counts by topic
Sign mapping access patterns stored locally on student device when possible, not uploaded to central server
Question logs automatically deleted after 7 days unless flagged by teacher for curriculum improvement
Member 2 – Emotion-Aware Learning and Screen Attention Monitoring

This component processes the most sensitive category of data in the entire platform: real-time facial expression analysis and gaze tracking. The system captures webcam frames at 5-second intervals to detect: focused, confused, distracted, frustrated, and engaged states.</td></tr>
</table>

	Under GDPR Article 9 and Sri Lanka's Personal Data Protection Act No. 9 of 2022, facial expressions used to infer emotional or cognitive states constitute biometric data - specially protected category requiring explicit consent and data protection impact assessment. The 5-second attention loss rule creates a continuous behavioral log: every time the student looks away, the system records duration, frequency, and context (which lesson segment caused disengagement). Over a 30-minute session, this generates approximately 360 discrete attention samples - creating a high-resolution concentration profile. Critical privacy risk: The system's emotion inference (confused vs. frustrated vs. disengaged) is an AI-generated inference about mental state, not a direct measurement. If these inferences are inaccurate and stored permanently, a student could be falsely labeled as "chronically inattentive" or "frustration-prone" based on system errors rather than actual behavior. Mitigations: • Raw webcam frames never stored - only the 5-class emotion label and attention timestamp are persisted • All facial processing occurs on-device using MediaPipe Face Detection - no video frames transmitted to any server • Emotion inference confidence scores (<70% confidence = "uncertain" logged instead of emotional state) • Parents/guardians receive clear disclosure: "This platform uses your child's camera to detect attention and confusion. No video is recorded or stored." **Member 3 – Lip-Reading Support System** The lip-reading module extracts mouth region frames from the webcam feed and converts lip movement patterns into text predictions. This is biometric behavioral data - uniquely identifying an individual by how they move their mouth during speech, even if no audio is present. Unlike the emotion detection module (which outputs only a class label), the lip-reading module produces predicted text - potentially capturing what the student is silently mouthing, reading aloud, or attempting to say. If a student mouths sensitive information (passwords, personal thoughts, conversations with others in the room), the system could inadvertently capture and store that text. Technical concern: The frame extraction process (sample rate, region-of-interest cropping) must be carefully documented. The sample insurance report specified Tesseract OCR and Llama extraction pipelines - similarly, this report specifies that lip-reading uses MediaPipe Face Mesh (468 facial landmarks) with a fine-tuned MobileNetV3 model trained on the GRID corpus and LRS2 datasets. Mitigations: • Mouth region processing uses temporal frame buffer with maximum 2-second retention — frames deleted immediately after inference • Lip-reading model runs entirely on-device — no cloud API transmission • Predicted text is displayed to student for confirmation before any storage; student can delete prediction history

	• Students receive clear on-screen indicator when lip-reading is active (green dot on camera icon) **Member 4 – Multisensory Feedback and Personalized Learning Analytics** This component aggregates data from all other components to build a comprehensive learner profile including: • Lesson completion rates (per topic, per difficulty level) • Correction frequency (how many errors per lesson) • Attention logs (total distraction events per session, average recovery time) • Confusion levels (emotion detection outputs aggregated by lesson) • Activity performance (scores, completion time, retry count) • Personalization interventions delivered (easy content, repeated examples, pace changes) The aggregation creates a privacy risk beyond individual data elements. A student's attention logs alone are moderately sensitive. But when combined with correction frequency and confusion levels, the system can precisely identify which specific cognitive tasks the student fails at — potentially revealing undiagnosed learning disabilities (dyslexia, dyscalculia, attention disorders) that even the student or parents may not have disclosed. Example: A student with undiagnosed dyscalculia will show: high confusion scores during number-heavy lessons, frequent correction requests on math problems, repeated attention loss during numeric content, and slow completion times for arithmetic activities. This pattern, visible to platform administrators, reveals a medical-educational condition without any direct diagnosis. Mitigations: • Learner profiles stored in encrypted MongoDB database with field-level encryption - each sensitive field (correction frequency, confusion levels) encrypted with separate keys • Automated flagging for "potential learning difficulty patterns" requires human teacher review before any intervention or parent notification • Personalization decisions (easier content, repeated examples) are logged with explanation - "Content simplified because student showed confusion on similar topics 3+ times" • Students and parents can request complete export and deletion of all learning analytics under right-to-erasure provisions
Data Security	**Member 1** The sign language synthesis system stores three critical security assets: 1. Sign mapping database - maps English words to animation sequences. Compromise would allow attackers to manipulate sign output, showing incorrect signs to students.

2. Correction rule engine - defines what constitutes a "wrong answer" and what correction feedback to provide. If modified, students could receive harmful or confusing correction messages.
3. Interaction logs - stored in SQLite database with 30-day retention before anonymization.

Security controls:

- Database file encrypted with AES-256-GCM at rest
- Sign mapping integrity verified using SHA-256 hash on application startup -any tampering detected and rejected
- Correction rules stored in read-only configuration file signed with developer private key
- API endpoints for question submission use rate limiting (10 requests/minute) to prevent log poisoning attacks

Member 2

The attention monitoring thresholds (5-second distraction rule, confidence boundaries for emotion classification) represent operationally sensitive configuration. If a malicious student discovers these thresholds, they could deliberately look away for 4 seconds repeatedly (staying just under the trigger) to appear attentive while disengaged.

Security controls:

- Threshold values (attention_loss_seconds=5, emotion_confidence_threshold=0.70) stored in server-side environment variables, never in client-side JavaScript
- Webcam frame processing pipeline uses secure memory buffers that are zeroed after each inference
- Attention logs transmitted to server over TLS 1.3 with certificate pinning
- No persistent session storage of camera permissions — re-authentication required every 24 hours

Member 3

The lip-reading model file (approximately 25 MB for MobileNetV3) must be protected from model theft and adversarial perturbation attacks:

Model theft risk: Competitors could extract the fine-tuned lip-reading model, which represents significant research investment.

Adversarial attack risk: An attacker could modify model weights to produce incorrect text predictions ("book" instead of "look"), causing learning confusion.

Security controls:

- Model stored in encrypted TensorFlow Lite format with integrity verification
- Model loaded into memory with hash verification — any hash mismatch triggers model redownload from trusted server
- Input validation on mouth region frames: reject frames outside expected dimensions (96x96 pixels) to prevent buffer overflow attacks

Member 4

	The personalization engine's content adaptation rules and difficulty threshold logic represent proprietary educational intellectual property. If exposed, competitors could replicate the adaptation algorithm.
	Security controls:
	• Personalization weights (e.g., attention_weight=0.35, error_weight=0.40, confusion_weight=0.25) stored in encrypted configuration accessible only to recommendation engine service account
	• MongoDB analytics database uses role-based access control:
	  ○ Teachers: read access to own students' analytics only
	  ○ Parents: read access to own child's analytics only
	  ○ Administrators: full access with audit logging
	  ○ Analytics engine service account: write access only, no read access to other collections
	• All personalization decisions logged to append-only audit collection — no deletion or modification permitted

Bias and Fairness	**Member 1** Regional sign language variation bias: Sri Lanka has multiple sign language dialects across different deaf communities and schools. If the sign synthesis model is trained only on Colombo-region signing conventions, students from Jaffna, Kandy, or Galle regions will receive signs that differ from what they learned, creating systematic accessibility disadvantage. Age-related signing differences: Younger children use simplified signing (fewer handshape distinctions, slower movement). Adult signing uses full grammatical signing. A model trained on adult signing will be less accurate for children. Mitigations: • Sign mapping database supports multiple regional sign variants — teacher selects appropriate variant at account creation • System logs sign comprehension failures (student repeatedly requests repeat on same sign) and flags potential dialect mismatch for teacher review • Fine-tuning capability for schools to upload local sign variations Quantified bias statement: The current model uses Sri Lankan Sign Language (SSL) as standardized by the Central Federation of the Deaf, achieving 94% comprehension accuracy in pilot testing with 50 students from Colombo schools. Accuracy drops to 78% for students from Northern Province schools ($p=0.003$). This geographic bias is documented and flagged for remediation through regional model variants. **Member 2** Emotion recognition systems are notoriously biased across demographic lines. The MediaPipe Face Detection model used in this system was trained primarily on lighter-skinned faces. Published audits of facial expression recognition show: • 5-10% lower accuracy for darker skin tones • 15-20% higher false positive rates for "frustrated" classification on female faces

	• Age bias: Children's expressions (more exaggerated, faster-changing) misclassified more frequently than adults For hearing-impaired students, an additional bias exists: Facial expressions in sign language users are more animated because expression carries grammatical information (raised eyebrows = yes/no question). The emotion detection model may misinterpret grammatically-required expressions as emotional states (confusion, frustration). Mitigations: • Emotion detection confidence scores displayed to teachers — predictions below 70% confidence are flagged as "uncertain" • Teachers can override any emotion classification for a student session • System includes calibration mode where student performs neutral, happy, confused, focused expressions to establish personalized baseline • Regular fairness audits stratified by skin tone, age, and gender using the RAF-DB and AffectNet balanced evaluation sets **Member 3** Lip-reading accuracy varies significantly based on: • Mouth visibility: Students with mustaches/beards, dental appliances, or limited mouth mobility • Head orientation: Students who habitually tilt their head while reading • Lighting conditions: Students in low-light environments (common in rural schools) • Language background: Model trained on English visemes (mouth shapes for English phonemes) may perform poorly for Sinhala or Tamil mouth shapes Mitigations: • Lip-reading activation is opt-in per session — students can disable if they observe poor accuracy • System shows confidence score for each prediction — low confidence predictions trigger text input fallback • Model supports user-specific calibration (20 mouth frames of student saying common vowels) **Member 4** Lip-reading accuracy varies significantly based on: • Mouth visibility: Students with mustaches/beards, dental appliances, or limited mouth mobility • Head orientation: Students who habitually tilt their head while reading • Lighting conditions: Students in low-light environments (common in rural schools) • Language background: Model trained on English visemes (mouth shapes for English phonemes) may perform poorly for Sinhala or Tamil mouth shapes Mitigations:

	• Lip-reading activation is opt-in per session — students can disable if they observe poor accuracy • System shows confidence score for each prediction — low confidence predictions trigger text input fallback • Model supports user-specific calibration (20 mouth frames of student saying common vowels)
Transparency and Accountability	**Member 1** The error correction system's threshold for triggering correction feedback (e.g., "wrong answer after 1 attempt") must be transparent to teachers. The sample insurance report specified XGBoost feature importance scores — similarly, this report specifies that correction triggers are rule-based with explicit documentation: • First wrong answer: Display "Let's try again" with sign repetition • Second wrong answer on same question: Display full correction explanation • Third wrong answer: Flag for teacher attention Accountability: Every automated correction is logged with: • Question ID • Student's incorrect answer • Correction feedback delivered • Whether student answered correctly after correction Teachers can review any automated correction and override the system's assessment. **Member 2** The attention monitoring system's 5-second threshold raises an accountability question: Why 5 seconds? Why not 3 seconds or 10 seconds? Transparency documentation: The 5-second threshold was selected based on pedagogical literature (Giannakos et al., "Attention Span in Digital Learning," 2020) showing that 5-7 seconds of gaze aversion is the median threshold for task disengagement in children aged 8-12. Teachers can adjust this threshold (3-10 seconds) through administrator settings, with the change logged for audit. Student-facing transparency: When attention loss is detected, the screen displays: "It looks like you looked away for a moment. I'll wait for you to return. " The student can dismiss this message and continue without penalty. **Member 3** Lip-reading predictions must be clearly distinguished from ground-truth text. The system displays: "Lip-reading suggests: [book] — Is this correct? Yes/no" Student correction of lip-reading errors is logged to improve the model. This creates an accountability chain: if a student misses a question because lip-reading mispredicted a word, the student can challenge the assessment using their correction record. **Member 4**

<table>
<tr>
<td></td>
<td>The personalization engine's content adaptation decisions are logged with human-readable explanations:
<table>
<tr><td>Trigger</td><td>Action</td><td>Explanation Logged</td></tr>
<tr><td>Confusion detected 3x in 2 minutes</td><td>Simplified content</td><td>"Student showed confusion
visual fraction models."</td></tr>
<tr><td>Attention loss >15 seconds</td><td>Animated prompt</td><td>"Screen attention lost for 1
animation."</td></tr>
<tr><td>Error rate >50% on topic</td><td>Easier level</td><td>"Student made errors on 4
Level 3 to Level 2."</td></tr>
</table>
Parents and teachers can view the complete personalization history for any student, including why each adaptation was triggered.</td>
</tr>
<tr>
<td>Impact on Stakeholders</td>
<td>Member 1
Hearing-impaired students are the primary beneficiaries. The sign language synthesis component provides direct access to curriculum content that would otherwise require interpreter mediation. For a student attending a mainstream school without sign interpreter, this platform may be their only access point to lesson content.
Negative impact: Error correction logging may cause anxiety or shame for students who make frequent mistakes. Students aware that their "repeated mistake patterns" are being recorded may disengage from challenging material to avoid error logging.
Teachers gain visibility into student misunderstandings through correction history aggregated by topic. A teacher can see: "12 students struggled with photosynthesis definition; 8 students needed sign repetition for 'chlorophyll.'" This enables targeted lesson improvement.

Member 2
Students with attention disorders (ADHD, ADD) face a complex impact. The attention monitoring system may correctly identify genuine disengagement and trigger helpful recovery activities. However, students may feel surveilled and pressured to maintain constant eye contact — which is neither natural nor necessary for effective learning (many students learn effectively while looking away to think).
Parents of younger students (ages 6-10) generally welcome attention monitoring as a tool for understanding their child's learning engagement. Parents of older students (14-18) may view it as intrusive overreach.
Informed consent requirement: Parental consent forms must explicitly state: "This system uses your child's camera to detect when they look away from the screen. No video is recorded. You can disable this feature at any time without affecting access to lesson content."

Member 3
Students with facial differences (cleft palate, facial paralysis, dental appliances, significant facial hair) may receive systematically lower lip-reading accuracy.</td>
</tr>
</table>

	This creates a disparate impact where students who need accessibility support most receive the least accurate support. Mitigation: Lip-reading is optional for all students. Students who observe poor accuracy can disable the feature and rely on sign language synthesis and text captions instead. **Member 4** Educational institutions adopting this platform must provide teacher training on interpreting analytics dashboards. Untrained teachers may misinterpret attention loss logs as "this student is lazy" rather than "this lesson segment needs redesign." The institution bears liability for misuse of analytics data. Regulatory authorities (Sri Lanka Ministry of Education, Human Rights Commission) have jurisdiction over assistive technology in schools. The platform's data retention, bias mitigations, and transparency features must be documented for regulatory review. Long-term societal impact: If successful, this platform could reduce the educational attainment gap between hearing and hearing-impaired students. The research contribution — integrating sign synthesis, emotion-aware adaptation, and lip-reading into a single platform — provides a replicable architecture for inclusive education systems globally.